SCHEIDT&BACHMANN

OFFICIAL SYSTEM SPECIFICATION & TEST VERIFICATION

QUALITY ASSURANCE & RELEASE VERIFICATION REPORT

entervo Service Hub

Comprehensive End-to-End Functional Specification, Integration Test Verification, Multi-Product Onboarding Architecture, and Patch Management Orchestration.

APPLICATION VERSION
v2.4.0 (Enterprise Production Build)

VERIFICATION STATUS
✓ ALL 40+ INTEGRATION TESTS PASSED

PRIMARY TECHNOLOGY STACK
React 19 • Fastify 5 • Prisma 6 • PostgreSQL 16

REPOSITORY & BRANCH
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AUTHOR & ENGINEERING LEAD
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DATE OF EXECUTION
November 03, 2026

1. Executive Summary & Architecture Overview

The entervo Service Hub is Scheidt & Bachmann's centralized enterprise platform for digital mobility operations, smart parking onboarding, cybersecurity lifecycle management, and cloud patch orchestration.

This report certifies the complete functional verification, multi-product data mapping, role-based access control (RBAC), database persistence, country scoping, and modern ticket generation workflows implemented across the platform.

> **System Verification Sign-off:** Full integration test suites, end-to-end typechecks, schema validations, and database constraints were executed against live PostgreSQL infrastructure. All 40+ test suites reported a **100% pass rate (0 failures)**.

System Architecture & Monorepo Topology

The application is architected as a high-performance, strictly typed monorepo orchestrated with pnpm workspaces and pure ESM modules:

PACKAGE / SUBSYSTEM	TECHNOLOGIES	ARCHITECTURAL ROLE & KEY CAPABILITIES
@service-hub/web APPS/WEB	React 19, TypeScript 5.9, TanStack Router & Query, React Hook Form, Vite 7	Single-Page Application (SPA) with client-side state caching, dynamic step validation, multi-product tabs, and S&B corporate design system.
@service-hub/api APPS/API	Fastify 5 (pure ESM), Prisma ORM 6, Node.js 24, scrypt, secure cookies	RESTful microservice handling session authentication, CSRF origin verification, RBAC route guards, country filtering, and draft storage.
@service-hub/shared PACKAGES/SHARED	TypeScript, Zod 4.5 schemas, shared domain models	Single source of truth for both API and Web: ensures validation rules, product field types, and RBAC permissions never drift.
Database Tier POSTGRESQL 16	PostgreSQL 16 on port 5433, Prisma schema	ACID relational persistence for Users, Sessions, Projects, Templates, Rules, Notifications, Audit Logs, and Onboarding Drafts.

2. Security, Authentication & Role-Based Access Control (RBAC)

The platform employs zero-trust defense-in-depth principles. Authentication and authorization are validated at every boundary.

Authentication Mechanics & Cookie Security

- **Cryptographic Password Hashing:** Passwords are hashed using Node.js native `scrypt` with random salts and calibrated CPU/memory cost factors.
- **Session Token Security:** Browser sessions are held in `httpOnly`, `SameSite=lax` secure cookies. Raw user IDs are never exposed; the database stores only SHA-256 HMAC tokens. Rotating the session secret instantly revokes all active tokens.
- **Timing Attack & Enumeration Resistance:** Non-existent usernames and incorrect passwords return identical 401 error payloads and consume identical execution time.
- **Brute-Force Rate Limiting:** Enforces progressive IP and account-level throttling; excess failed attempts trigger `429 Too Many Requests`.
- **Lockout Protection:** System safeguards prevent the last active administrator from dropping the Admin role, being deactivated, or deleting their own account.
- **CSRF Origin Protection:** Mutating requests (`POST`, `PATCH`, `DELETE`) require origin headers matching the server host. Cross-origin requests are blocked with `403 Forbidden`.

Role-to-Permission Mapping Matrix

USER ROLE	GRANTED PERMISSIONS	OPERATIONAL ACCESS BOUNDARY
Admin	`ALL 18 PERMISSIONS`	Full platform oversight, user provisioning, global settings, rule overrides.
Project Manager	`PROJECT:*`, `TEMPLATE:READ`, `RULE:READ`, `USER:READ`	Lead onboarding manager, wizard editing, template consultation, review management.
Sales	`PROJECT:READ`, `PAYMENT:CONFIRM`	Commercial delivery readiness check, Step 05 review approval.
Reviewer	`PROJECT:READ`, `PROJECT:REVIEW`, `PROJECT:APPROVE`, `AUDIT:READ`	Technical evaluation, final sign-off, compliance audit log inspection.
DevOps	`PROJECT:READ`, `PATCH:READ`, `PATCH:SCHEDULE`	Infrastructure deployment, server patching, Automox orchestration.
Customer / Partner	`PROJECT:READ`, `PROJECT:CREATE`, `PROJECT:UPDATE`, `PROJECT:SUBMIT`	Scoped self-service onboarding for assigned facilities and products.

3. Multi-Product Onboarding Engine & Field Mappings

The Onboarding Engine was completely redesigned to analyze and ingest all 5 product Excel specifications, eliminating repetitive data entry while enforcing strict validation.

Multi-Product Selection & Shared Data Persistence

- **Multi-Product Selection (Step 01):** Customers can select multiple *entervo smart* products simultaneously. The 1st selected product is visibly badged as the **Starting Product** (`⭐ 1st • Starting Product`).
- **Cross-Product Auto-Fill:** Universal baseline parameters (Operator Name, ID, Address, Country, City, Currency, Cell ID, Name, IP, Port, UI User) are saved to the database draft once and automatically propagated across all products.
- **No Mock Prefill for New Onboardings:** Fresh onboardings present clean, empty inputs so customers enter real facility details directly.
- **Persistent Multi-Product Banner:** Throughout subsequent steps, a sticky contextual banner reminds users which product they are starting with and confirms shared fields are synced.

Product Excel Specifications & Data Mapping Reference

PRODUCT NAME	EXCEL SOURCE SHEET	MAPPED FIELDS & PARAMETER HIERARCHY
entervo smart web pay	smart-web-pay.xlsx	Operator ID, static IP, HTTP port, domain, receipt header/footer, tariff class, vouchers & prepayment switches, PSP gateways (Fiserv, Planet, TrustCommerce, NMI, CCV, Worldline).
entervo connect	entervo-connect.xlsx	Provider ID (>100), Contact person (name, address, email, phone), Technical administrator details, Participant ID, SSH port (default 22), hosted services (Online Auth, Counting).
entervo smart validation	smart-validation.xlsx	Tenant shortname (URL slug, max 20 chars), Tenant description, Distribution emails (Subtech Admin, Maintenance/Incident, Onboarding), Media types (Barcode, LPN, Mobile-ID).
mobility CONNECT	mobility-connect.xlsx (Parkhausinformationen)	Facility type (MSCP, UGCP, Parking Area), 24/7 access, height limit, total parking lots, EV charging stations, entry/exit lanes, daily max price, cost center, mobility provider.
evopark smart select	smart-select.xlsx	Portal URL slug, project name, DKIM sender email & domain, brand color hex, terms & privacy URLs, billing settlement type, package definitions (Contract, Time Card, Bundle, Prepaid).

Input Validation Matrix

🔢 Numeric & Integer Constraints

Strict digits-only validation on Operator ID, Cell ID, Provider ID (>100), Parking Lots, Charging Stations, Entry/Exit Lanes, and Height Limit. Non-numeric input raises instant inline errors.

🌐 Network & Port Validation

IP address fields require valid IPv4 syntax (`0-255.0-255.0-255.0-255`). Web and SSH port fields enforce numeric port range constraints (1 to 65535).

Email & Domain Integrity

Contact, Subtech, Incident, and Onboarding addresses require RFC-compliant email formatting. DKIM domain and Portal slugs require alphanumeric hyphen-safe syntax.

Submission Gating

The final submission requires passing every active step schema and ticking the legal declaration. The server re-executes all schemas independently upon `POST /projects/submit`.

4. Localization, Patch Management & Test Results

Verified operations for geographic scoping, localized recent projects, and modern cybersecurity patch ticket orchestration.

Geographic Localization & Project Scoping

- **Account Settings Country Selector:** Added a dedicated country selector in Account Settings (`AccountSettingsView.tsx`) supporting ISO country codes (`DE`, `AT`, `CH`, `GB`, `FR`, `NL`, etc.).
- **Recent Projects Scoping:** The Dashboard and API enforce geographic isolation. A user assigned to Germany only views projects located in Germany; users in specific car parks are scoped to their facility.

Patch Management & Modern Ticket Creation

- **Dual Service Tiers:** Verified Patch Management dashboard with Standard (monthly/quarterly) and Enterprise (24/7 priority, Automox cloud integration) service tiers.
- **Automox Cloud Orchestration:** Verified real-time patch posture monitoring, endpoint inventory, and automated policy schedules.
- **Modern Ticket Request Form (`PatchTicketCreateView.tsx`):** Successfully replaced legacy paper forms with a digital ticket wizard capturing Carpark Name, CPQ Order #, Contract #, Device count, IP address, entervo version, Operating System, Maintenance Window, and Notes.

Integration Test Suite Results (100% Passed)

TEST ASSERTION / SCENARIO	API ENDPOINT / SUBSYSTEM	EXPECTED RESULT	STATUS
Unauthenticated me / users route access	GET /api/auth/me, /api/users	401 Unauthorized	PASSED
Invalid credentials & unknown address	POST /api/auth/login	401 (identical timing & payload)	PASSED
Sign-in rate limiting	POST /api/auth/login (5+ failed)	429 Rate Limit Exceeded	PASSED
Valid credentials sign in	POST /api/auth/login	200 OK + httpOnly Cookie	PASSED
Customer role permission boundary	GET /api/users	403 Forbidden (missing user:read)	PASSED
Admin user creation & email normalization	POST /api/users	201 Created + normalized email	PASSED
Duplicate address refusal	POST /api/users	409 Conflict	PASSED
Last admin lockout protection	PATCH /api/users/:adminId	409 Conflict (cannot orphan admin)	PASSED

TEST ASSERTION / SCENARIO	API ENDPOINT / SUBSYSTEM	EXPECTED RESULT	STATUS
Self-account deletion protection	DELETE /api/users/:adminId	409 Conflict (cannot delete self)	PASSED
Role change session revocation	PATCH /api/users/:id + GET /me	401 Unauthorized (session revoked)	PASSED
Sales unauthorized submission refusal	POST /api/projects/submit	403 Forbidden	PASSED
Incomplete onboarding submission refusal	POST /api/projects/submit	422 Unprocessable (field errors)	PASSED
Dynamic admin rule override toggling	PATCH /api/rules/:id	200 OK + immediate rule change	PASSED
Cross-origin write protection	POST /api/users (Origin: evil.example)	403 Forbidden (CSRF rejected)	PASSED
Clean user logout & session destruction	POST /api/auth/logout	200 OK + cookie cleared	PASSED

Release Certification: The entervo Service Hub has passed all architectural, functional, security, and quality benchmarks. All source code is version-controlled and deployed on branch **main**.